

Experimental evidence on the productivity effects of generative AI (Noy & Zhang, 2023)

Read the whole thing

Reading Notes

not peer reviewed

the question with automation is always whether humans will be displaced or complemented

put another way, if you can do something more quickly (or just more), does that increase demand?

and there's always the question about whether it affects low-ability or high-ability workers and in what ways

in their intervention (ChatGPT for writing tasks) the entire curves shifted, with treatment groups taking less time to produce better results

low-ability (those who scored low on the initial task) participants gained more from the use of ChatGPT than high-ability participants (those who scored high on the initial task)

participants using ChatGPT primarily seemed to use it to replace their work, doing only minimal editing after pasting in a ChatGPT-generated response

more time spent active after pasting in the ChatGPT response was not correlated with increased scores on the evaluation (!)

they found no evidence that human editing improved the ChatGPT output (!!)

both job satisfaction and self-efficacy increased with the use of ChatGPT

LLMs are often limited in their context-/domain-specific knowledge, which limits the set of tasks that they can easily be set to

Reading Questions

1. What type of tasks were used in the Noy & Zhang experiment? Are these tasks representative of the most cognitively demanding work in your professional role?

They used fairly basic writing tasks such as drafting e-mails and press releases. They are not representative of the most demanding work in my role.

2. The study found that AI 'compressed' the skill distribution — lower-skilled workers improved more. What are the institutional implications of this finding for how AI tools are deployed in higher education staff contexts?

For staff and faculty, it could mean that less onboarding and training is needed for new/junior employees to reach a certain threshold of proficiency. This would also be true for students - they might be able to develop skills in a shorter amount of time. However, in both cases, I would be concerned about skill atrophy (for already-proficient workers) or non-skill development in non-proficient workers and students since some of the task would be offloaded to AI.

3. The paper found quality improvements in addition to time savings. What metric was used to assess 'quality,' and is that metric applicable to the documents you produce?

The metric used to assess quality was a qualitative assessment (grade) by a panel of peers. I think this would be appropriate for documents that I produce, although that would only be used for either research or other inquiry purposes.

4. What is one methodological limitation of this study that would affect your confidence in applying its findings to your specific work context?

This isn't methodological *per se*, but the study wasn't peer reviewed, which makes me skeptical of their findings.

Additionally, as the authors note, their choice of tasks which are well-suited for ChatGPT and other LLMs in some ways represents a best-case scenario/upper bound for how much AI might help to improve productivity.

Key Terms

Term	My Definition
Randomized controlled trial (RCT)	a style of research study in which participants are sorted by chance into different conditions, usually control and treatment (possibly with multiple treatment conditions)
Treatment effect	the observable changes that occur (ostensibly) as a result of a treatment condition in an experiment
Skill compression	this term isn't used in the text, but I would guess that it might refer to the ability of AI to allow workers to perform tasks/produce outputs that fall along a narrower, higher distribution of quality

Task completion time	how long it takes a person to complete a given task
Output quality assessment	judgments of how good a human- or AI-created product is

Navigating the jagged technological frontier: Field experimental evidence of the effects of AI on knowledge worker productivity and quality (Dell'Acqua et al., 2026)

Read Abstract, Section 1 (Introduction), Section 2 (Experimental Design), Section 4 (Results), and Section 6 (Discussion) (~20 pages total).

Reading Notes

jagged frontier = uneven impact of technology such as AI such that it makes some tasks easier/faster while making some harder/worse/slower

it's not easy for workers to discern whether a particular task is within or beyond the jagged frontier, and this is important for the future of human-AI collaboration - this is reinforced by how confident LLM-produced content sounds

broadly, they found that AI improves task performance as measured by outsiders

using ChatGPT + getting training showed the greatest impact, although using ChatGPT alone still resulted in high impact

as in Noy and Zhang (2023), lower-skilled subjects benefited most from the treatment

AI assistance decreased performance on a beyond-the-frontier task, although it increased the coherence and persuasiveness of those using it

basically, AI's great for inside-the-frontier tasks but lousy for outside-the-frontier tasks (as one might expect)

they think that their findings are not as applicable in situations where you're trying to optimize for speed (customer service, manufacturing, etc.)

important to evaluate whether a task is inside the frontier or outside it, though they don't say anything about how to do that

prior literature suggests that AI helps more with ideation and content creation than making decisions

in addition to being jagged, the frontier is not fixed

Reading Questions

1. What is the 'jagged frontier' and why is it 'jagged' rather than a smooth capability boundary?

The jagged frontier refers to AI's ability to perform well or perform poorly on different tasks that could be regarded as equivalent in difficulty. This arises from the peculiarities of how AI works and how humans function and work differently from AI.

2. How did Dell'Acqua et al. identify which tasks were 'inside' vs. 'outside' the frontier experimentally? What implications does this classification method have for practitioners who cannot run experiments?

They used their own judgment from their experience in working with AI to decide which tasks were either inside or outside the frontier. However, as they note in a few places in the paper, non-experts will likely struggle to discern in the same way, so caution and testing will be needed for non-experts.

3. What does the 'centaur' collaboration style mean in this paper? What does the 'cyborg' style mean? Which is more appropriate for the task you will complete in Task 2?

Neither of these terms are defined in the paper (they appear to come from Randazzo et al., 2025, cited in the References section). The “centaur” style seems to refer to discretely defining portions of a workflow to be performed by a human or by an AI, while the “cyborg” style seems to refer to integrating AI into all portions of a workflow. For Task 2, I think a centaur model is likely more appropriate because I’m just getting started with AI integration, so a clearer delineation likely makes sense. I don’t think I’d be ready to integrate AI at every phase just yet.

4. The paper found performance degradation for AI-assisted workers on outside-frontier tasks. What cognitive mechanism do the authors propose to explain this? Do you find the explanation convincing?

They suggested that people were relying too much on their AI tools and unwilling or unable to recognize that it was giving them unhelpful input. This seems reasonable to me since LLMs tend to sound very confident in their output, and we’re used to equating confidence with correctness.

Key Terms

Term	My Definition
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Jagged technological frontier	the idea that there are some tasks that are within the capability of a technology (e.g., AI) to perform, while there are some that are outside its capability, despite their seeming to be of equivalent difficulty
Centaur collaboration	not defined in this paper, but refers to dividing AI and human works into discrete parts/steps
Cyborg collaboration	not defined in this paper, but refers to the integration into all portions of a workflow
Over-reliance	depending too much on something (e.g., AI)
Inside-frontier task	a task that an AI tool can perform well and/or aid a human in performing
Outside-frontier task	a task that an AI tool cannot perform well and/or will hinder a human in performing